fred

Part C

Course title: SC7 Bayes Methods
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1. Throughout this question $q(\theta'|\theta) = q(\theta|\theta')$ is a symmetric proposal density.

(a) [15 marks]

(i) Let $Y \sim p(\cdot|\theta)$, $Y \in \mathcal{Y}$ and let $\theta \in R$ be a parameter with prior $\pi(\theta)$ and, for data $Y = y_{obs}$, posterior $\pi(\theta|y_{obs})$. Explain what it means for $\pi(\theta|y_{obs})$ to be *doubly intractable* and the consequences of this for MCMC targeting $\pi(\theta|y_{obs})$.

(ii) Consider the following MCMC-ABC algorithm with summary statistics $S(y) \in R^p$ for $y \in \mathcal{Y}$, distance $d(S, S')$ and threshold δ:

Suppose $X_t = \theta$.
Step 1. Simulate $\theta' \sim q(\cdot|\theta)$ and $y \sim p(\cdot|\theta')$.
Step 2. If $d(S(y), S(y_{obs})) < \delta$ then accept θ' (set $X_{t+1} = \theta'$) with probability

$$\alpha(\theta'|\theta) = \min\left\{1,\ \pi(\theta')/\pi(\theta)\right\}$$

and otherwise reject θ' (set $X_{t+1} = \theta$).

Show this algorithm satisfies detailed balance for $\pi(\theta|d(S(Y), S(y_{obs})) < \delta)$.

(iii) Let $\theta \sim \pi(\cdot|y)$ and $\theta' \sim \pi(\cdot|y_{obs})$. Suppose the following conditions hold.

1. The posterior mean $\mu(S(y)) = E(\theta|Y = y)$ is a linear function of $S(y)$ alone, that is, if $s = S(y)$ and $s_{obs} = S(y_{obs})$ then, for some $\beta \in R^p$,

$$\mu(s) = \mu(s_{obs}) + (s - s_{obs})^T\beta.$$

2. The random variables $\theta - \mu(s)$ and $\theta' - \mu(s_{obs})$ are identically distributed.

Let $\theta_{adj} = \theta - (s - s_{obs})^T\beta$. Show that $\theta_{adj} \sim \pi(\cdot|y_{obs})$ and explain how to use this result to make a regression adjustment to the MCMC-ABC output.

(b) [10 marks] Consider the following variant of MCMC. Let $\xi(x|\theta, \theta')$ be a density for real x and let $f(x)$ be a differentiable involution, so $f(f(x)) = x$. Let $f' = df/dx$.

Suppose $X_t = \theta$.
Step 1. Simulate $\theta' \sim q(\cdot|\theta)$ and $x \sim \xi(\cdot|\theta, \theta')$.
Step 2. Accept θ' (set $X_{t+1} = \theta'$) with probability

$$\alpha(\theta'|\theta; x) = \min\left\{1,\ \frac{\pi(\theta'|y_{obs})\xi(f(x)|\theta', \theta)}{\pi(\theta|y_{obs})\xi(x|\theta, \theta')}|f'(x)|\right\}$$

and otherwise reject θ' (set $X_{t+1} = \theta$).

(i) Show that this algorithm satisfies detailed balance with respect to $\pi(\theta|y_{obs})$.
[*Hint: What is the marginal probability $\alpha(\theta'|\theta)$ to accept? Also, $f'(x) = 1/f'(f(x))$.*]

(ii) Suppose $p(y|\theta)$ is intractable but we can estimate

$$D(\theta, \theta') = \log\left(p(y_{obs}|\theta')/p(y_{obs}|\theta)\right)$$

with $\widehat{d} \sim N(D(\theta, \theta'), \sigma^2)$. Show that the following algorithm targets $\pi(\theta|y_{obs})$.

Suppose $X_t = \theta$.
Step 1. Simulate $\theta' \sim q(\cdot|\theta)$ and $\widehat{d} \sim N(D(\theta, \theta'), \sigma^2)$.
Step 2. Accept θ' (set $X_{t+1} = \theta'$) with probability

$$\alpha(\theta'|\theta; \widehat{d}) = \min\left\{1,\ \exp(\widehat{d} - \sigma^2/2)\frac{\pi(\theta')}{\pi(\theta)}\right\}$$

and otherwise reject θ' (set $X_{t+1} = \theta$).

[*Hint: take $\xi(x; \theta, \theta') = N(x; 0, \sigma^2)$ and $f(x) = \sigma^2 - x$.*]

2. (a) [7 marks]

(i) Let S be a sample space and let $\mathcal{S}$ be a σ-field of sets in S. Write down the first three Savage axioms in the form given by de Groot.

(ii) Suppose $S = [0,1]$ and $\mathcal{S}$ is the set of all subsets $A \subseteq S$ such that $\int_A dx$ is defined. Suppose our preference is always for the larger interval, so if $A, B \in \mathcal{S}$ then $\int_A dx < \int_B dx$ if and only if $A < B$ with $\int_A dx = \int_B dx$ when $A \sim B$. Also, $\int_\emptyset dx = 0$. Show that these preferences satisfy the first three Savage axioms.

(b) [8 marks] Let $X_1, X_2, X_3, ...$ be an infinite exchangeable sequence of binary random variables.

(i) What does it mean to say that these variables are exchangeable?

(ii) State de Finetti's theorem for the sequence $X_1, X_2, X_3, ...$

(iii) Show that $\Pr(X_1 = x_1, X_2 = x_2, X_3 = x_3) = \Pr(X_{98} = x_1, X_{99} = x_2, X_{100} = x_3)$ for all $(x_1, x_2, x_3) \in \{0,1\}^3$.

(c) [10 marks] For $\theta \in R^p$ and $\psi \in R^q$ let $p_0(\theta) = q_0(\theta)/c_0$, $p_1(\theta,\psi) = q_1(\theta,\psi)/c_1$ and $p_2(\theta,\psi) = q_2(\theta,\psi)/c_2$ be probability densities with intractable normalising constants c_0, c_1 and c_2. Let $Q(\psi)$ be a probability density which is tractable and normalised. Let $r = c_0/c_2$. The following results are subject to regularity conditions which are omitted.

(i) Show that

$$r = \frac{E_{(\theta,\psi)\sim p_1}(q_0(\theta)Q(\psi)/q_1(\theta,\psi))}{E_{(\theta,\psi)\sim p_1}(q_2(\theta,\psi)/q_1(\theta,\psi))}$$

(ii) Let $(\theta^{(t)}, \psi^{(t)}) \sim p_1$ be independent for $t = 1, ..., T$. Consider the r-estimator

$$\widehat{r} = \frac{\sum_{t=1}^T q_0(\theta^{(t)})Q(\psi^{(t)})/q_1(\theta^{(t)},\psi^{(t)})}{\sum_{t=1}^T q_2(\theta^{(t)},\psi^{(t)})/q_1(\theta^{(t)},\psi^{(t)})},$$

and let the relative mean square error be defined as

$$RE(\widehat{r}) = \frac{E[(\widehat{r}-r)^2]}{r^2},$$

where the expectation is taken over the random samples $(\theta^{(t)}, \psi^{(t)})$, $t = 1, ..., T$. It may be shown (using the delta-rule) that

$$RE(\widehat{r}) = \frac{1}{T} E_{(\theta,\psi)\sim p_1}\left(\frac{(p_0(\theta)Q(\psi) - p_2(\theta,\psi))^2}{p_1(\theta,\psi)^2}\right) + o(T^{-1}).$$

Show that $RE(\widehat{r})$ is minimised to leading order in T over densities p_1 by the choice

$$p_1(\theta,\psi) \propto |p_0(\theta)Q(\psi) - p_2(\theta,\psi)|.$$

[*Hint: the Cauchy Schwarz inequality is* $\left|\int a(x)b(x)dx\right|^2 \leq \int |a(x)|^2 dx \int |b(x)|^2 dx$.]

3. (a) [13 marks] Let $\alpha > 0$ be a real parameter and H a base distribution on R. Let $\theta = (\theta_1, ..., \theta_n)$, $\theta \in R^n$. Let $G \sim \Pi(\alpha, H)$ be a Dirichlet Process (DP) and let $\theta_i \sim G$ independently for $i = 1, ..., n$. Let $\theta^* = (\theta_1^*, ..., \theta_K^*)$ give the unique values in θ and let $S = (S_1, ..., S_K)$ partition $[n] = \{1, 2, ..., n\}$ with $S_k = \{i \in [n] : \theta_i = \theta_k^*\}$ for $k \in [K]$.

Recall the Chinese Restaurant Process (CRP) realising (θ^*, S):

1. set $K = 1$, $S_1 = \{1\}$ *and* $S = \{S_1\}$

2. for $i = 1, ..., n-1$
with probability $\alpha/(\alpha + i)$ *do (a) and otherwise do (b):*
(a) set $S_{K+1} = \{i + 1\}$, $S \leftarrow S \cup S_{K+1}$ *and* $K \leftarrow K + 1$.
(b) simulate $k \sim (|S_1|, ...|S_K|)/i$ *and set* $S_k \leftarrow S_k \cup \{i + 1\}$.

3. simulate $\theta_k^* \sim H$ *independently for* $k = 1, ..., K$.

Let $P_{\alpha,[n]}(S)$ give the probability to output partition S for "customer" labels $[n]$.

(i) Calculate $P_{\alpha,[n]}(S)$ in terms of α, K and the set sizes $(|S_1|, ..., |S_K|)$.

(ii) Suppose $S \sim P_{\alpha,[n]}$. Take $\sigma \subset [n]$ and let $S^{(\sigma)} = (S_1^{(\sigma)}, ..., S_{K^{(\sigma)}}^{(\sigma)})$ be the partition we get if we run the CRP to get S and then remove any customers not in σ. Any sets left empty in S are removed, so $1 \le K^{(\sigma)} \le K$. Show that $S^{(\sigma)} \sim P_{\alpha,\sigma}$.

(iii) Let $n=100$ and $\alpha=2$. Give the probability 98, 99 and 100 are all in different sets.

(b) [12 marks] Let X be an $m \times n$ binary matrix with $X_{i,j} \sim \text{Bernoulli}(\theta_{i,j})$, $(i, j) \in [m] \times [n]$ all independent. Cluster the $\theta_{i,j}$-values in different "cells" (i, j): let $R = \{R_1, ..., R_K\}$ partition the row labels $[m]$ and for $k \in [K]$ let $C_k = \{C_{k,1}, ..., C_{k,L_k}\}$ partition the column labels $[n]$ in row-block R_k. For $k \in [K]$ and $\ell = 1, ..., L_k$ let

$$c_{k,\ell} = \{(i, j) \in [m] \times [n] : i \in S_k,\ j \in C_{k,\ell}\}$$

be the cells in column partition ℓ of row-partition k. Let $\theta_{k,\ell}^* \in [0, 1]$ be the probability parameter for all cells in $c_{k,\ell}$, so that $\theta_{i,j} = \theta_{k,\ell}^*$ for all cells $(i, j) \in c_{k,\ell}$.

(i) Let $C = (C_1, ..., C_K)$ and $\theta^* = (\theta_{k,\ell}^*)_{k\in[K],\ \ell\in[L_k]}$. The prior for the row partition is $P_{\alpha,[m]}(R)$. The column partitions have independent priors $P_{\beta,[n]}(C_k)$ and the probability parameters have independent priors $h(\theta_{k,\ell}^*) = \text{Beta}(\theta_{k,\ell}^*; u, v)$.

Give the posterior $\pi(R, C, \theta^*|X)$ in terms of the priors and observation model (up to a constant not depending on R, C, θ^*).

(ii) Calculate $\pi(R, C|X)$ (up to a constant) by integrating $\pi(R, C, \theta^*|X)$ over θ^*.

(iii) Let $Q \sim \Pi(a, \Pi(b, F))$ be a DP with a base distribution $\Pi(b, F)$ which is itself a DP with base distribution F. Let $G_i \sim Q$ independently for $i \in [m]$ and $G = (G_1, ..., G_m)$. Let $G^* = (G_1^*, ..., G_K^*)$ give the unique G_i's in G and let S be the corresponding partition of $[m]$. For $k \in [K]$ let $p_{k,j} \sim G_k^*$ independently for $j \in [n]$ and $p_k = (p_{k,1}, ..., p_{k,n})$. Let $p_k^* = (p_{k,1}^*, ..., p_{k,L_k}^*)$ give the unique $p_{k,j}$'s in p_k and let D_k give the corresponding partition of $[n]$. Let $D = (D_1, ..., D_K)$ and $p^* = (p_1^*, ..., p_K^*)$.

Show that (S, D, p^*) has the same distribution as (R, C, θ^*) for some suitable a, b and F which you should give.